

PAPER_I_EXT_AUDIT — 01_algebraic_identities

Script 01 — Algebraic Identity Verification (Gate E)

Audit: PAPER_I_EXT_AUDIT (Affine Profile Reduction for Fractional Triangle Packings in Split Graphs) · **Script:** 01_algebraic_identities.py

Environment: Python 3.14.4, sympy 1.14.0 (exact symbolic algebra)

Purpose

Re-derive, symbolically and exactly, every closed-form algebraic identity that carries proof weight in the manuscript / calculation ledger. These are the load-bearing steps behind the quadratic benchmark $R(p, q)$ and the conversion to n . Exact symbolic equality (LHS – RHS simplifies to 0) is proof-grade for these steps (**CALCULATION_VERIFIED**), independent of the manuscript's own algebra.

Symbols $p=|K|$, $q=|I \geq 2|$, $s=|S|$, $o=p-s$;

$A=s(s-1-q)/2$, $B=so$, $C=o(o-1)/2$; $U=(A+B+C)/3$, $D=A+C$, $H=A+(B+C)/3$;

$R(p, q)=(2p^2-2pq-q^2)/12$.

Identities checked (all EXACT, difference = 0)

Tag	Identity	Result
L48	$12(U-R) = q(2o+q) - 2p$	OK
L50	$12(D-R) = 12o^2 - 6o(2p-q) + (2p-q)^2 - 6p$	OK
L51	$12u^2 - 6uv + v^2 = 12(u-v/4)^2 + v^2/4$ (SOS ≥ 0)	OK
L53	$12(H-R) = (2s-q)^2 + 2q(p-s) - 2p - 4s$	OK
L58	$C(p, 2) - 2R + p = (p+q)^2/6 + p/2$	OK
$q=0$	$R(p, 0) - p/2 = (p^2-3p)/6$	OK
$q=0$	$M(0) - R(p, 0) = -p/6$ with $M(0)=p(p-1)/6$	OK
L25	$V_{\text{com}} = b \geq 2/2 + M$ (from $\Sigma(1-\kappa)=C(p, 2)-b \geq 2/2$)	OK
L28	$ E - 2V_{\text{com}} = C(p, 2) - 2M + b$	OK

9 / 9 identities exact.

Sign facts (inequality directions), on the domain $q, o, s \geq 0, s \leq p$

- $12(U-R)+2p = q(2o+q) \geq 0 \blacksquare U-R \geq -p/6 \geq -p/2$.
- $12(D-R)+6p = 12(o-(2p-q)/4)^2 + (2p-q)^2/4 \geq 0$ (verified equal to the SOS form)

■ $D-R \geq -p/2$.

• $12(H-R)+2p+4s = (2s-q)^2 + 2q(p-s) \geq 0$, and $-2p-4s \geq -6p$ since $s \leq p$

■ $H-R \geq -p/2$.

Hence each orbit candidate satisfies $\geq R(p, q) - p/2$, so $\min\{U, D, H\} \geq R - p/2$.

Verdict

ALL EXACT (CALCULATION_VERIFIED). The manuscript's algebra behind the quantitative bound and the final conversion contains no transcription or algebraic error on any of the checked identities. This is the exact-algebra half of Gate E; the LP-optimality half (that $M(\kappa^S) = \min\{U, D, H\}$) is checked numerically in Script 02.